3AL Test 2 2023

Time: 90 minutes **Available marks:** 65 **Full marks:** 60

Instructions

1. Answer all questions in the lined answer booklets.
2. You can answer the questions in any order but start each question on a new page.
3. Show all work and reasoning unless otherwise instructed.
4. This test consists of 4 questions. Make sure your question paper is complete.
5. If you have read the instructions, write your favourite Greek letter on the front cover of your answer booklet. You will receive an extra mark for doing so.

Good luck! ✧

1. Consider the group $\mathbb{Z}_2$ and the set $\mathbb{R}$. Construct a group action $*$ by $\mathbb{Z}_2$ on $\mathbb{R}$ such that $\mathbb{Z}_2 * x = \{x, -x\}$ for all $x \in \mathbb{R}$. Prove that $*$ is indeed a group action. Then find all stabilisers.
   [10 marks]

2. Let G be a group acting on a set X. Prove that if $x, y \in X$ lie in the same orbit, then $|S(x)| = |S(y)|$.
   HINT: Construct a bijection between $S(x)$ and $S(y)$.
   [15 marks]

3. (a) Let p be a prime. Suppose G is a p-group acting on a set X.
      (i) Write the definition of the fixed set, X_f.
      (ii) Write a statement (seen in lectures/the notes) that relates $|X|$ to $|X_f|$.
   (b) Let G be a p-group and X the set of subgroups of G.
      (i) Show that conjugation by G on X is a group action.
      (ii) Prove that p divides the number of non-normal subgroups of G.
   [3 + 2 + 5 + 10 = 20 marks]

4. Consider the following theorem and proof. Then answer the questions that follow.

Theorem. *Let p be a prime and G a non-abelian group such that $|G| = p^3$. Then G has $p^2 - 1$ nonsingleton conjugacy classes.*

Proof.

(1) Since $Z(G)$ is a subgroup of G, we have either $|Z(G)| = 1$, p, p^2 or p^3.

(2) However, it is not possible for $|Z(G)| = 1$ or p^3.

(3) If $|Z(G)| = p^2$, then $|G/Z(G)| = p$. Hence $G/Z(G)$ is cyclic and so G is abelian, a contradiction. Thus, $|Z(G)| = p$.

(4) According to the Class Equation,

$$p^3 = p + \sum_{i=1}^{n} |G : C_G(a_i)|,$$

where the a_is are (...).

(5) For each i, we must have $|G : C_G(a_i)| = 1$, p, p^2 or p^3.

(6) By how each a_i is defined, we know that $|G : C_G(a_i)|$ cannot be 1.

(7) If $|G : C_G(a_i)| = p^3$, then $C_G(a_i) = \{1\}$, which is not possible.

(8) If $|G : C_G(a_i)| = p^2$, then $|C_G(a_i)| = p$, which again is not possible.

(9) (...) $\square$

(a) Explain why (2) holds.

(b) Fill in the blank from (4).

(c) Write the definition for the centraliser of an element.

(d) Which theorem does (6) follow from?

(e) Explain why, in (7), it is not possible for $C_G(a_i) = \{1\}$.

(f) Explain why, in (8), it is not possible for $|C_G(a_i)| = p$.
HINT: How are $C_G(a_i)$ and $Z(G)$ related?

(g) Complete the proof, i.e. fill in the blank from (9).

[$4 + 4 \times 2 + 3 + 5 = 20$ marks]